

Power Query

30-Minute Hands-On Review | Week 9 - Section 3A

Learning Objectives: Explain what Power Query is and how it fits into Power BI | Describe how queries are structured as steps | Define query folding and explain its performance benefit | Identify tasks that are and are not a good fit for Power Query | Distinguish between Power Query Editor (GUI), M Language, and Dataflows

Instructions: Complete all four parts in 30 minutes. Refer to your Section 3A notes. Be ready to discuss your answers with the class.

Part 1: Fill in the Blank (~7 minutes)

Complete each sentence with the correct term or phrase from Section 3A.

1.	The underlying code language that Power Query uses to describe where data comes from and what transformations to apply is called _____.
2.	In Power Query, a query is broken into individual _____, each with a name and a specific transformation it performs.
3.	Because Power Query re-applies all steps automatically when data changes, it can be thought of as a _____ for data preparation.
4.	The technique where Power Query pushes transformation logic such as filtering or grouping back to the source database engine is called _____.
5.	A _____ decouples the data transformation layer from modeling and visualization, and runs Power Query in an application-agnostic way on the cloud.
6.	To write or modify M expressions beyond what the GUI supports, analysts use the _____ within Power Query Editor.

Part 2: Matching (~6 minutes)

Write the letter of the correct description next to each term.

Term / Concept	Description
A. Power Query Editor	1. Re-fetches the source data and re-applies all transformation steps in the pipeline
B. Query Folding	2. A cloud-hosted, application-agnostic use of Power Query that separates transformation from modeling
C. M Language	3. The underlying code automatically written as you interact with the Power Query GUI

D. Refresh Data	4. A single named operation in a query that performs one transformation (e.g., Remove Columns)
E. Power Query Step	5. The graphical user interface with wizards, dialogs, and ribbon tools for transforming data
F. Dataflow	6. Pushes transformation logic (filtering, grouping, joins) back to the source database to reduce data transfer

Part 3: Good Fit or Not? (~8 minutes)

For each use case, place a checkmark (✓) in the correct column. Use your Section 3A notes on Power Query strengths and limitations.

#	Use Case	Good Fit?	Not a Good Fit?
1	Extracting sales data from a SharePoint list and cleaning column names		
2	Merging a customer table with a regional lookup table before loading to the model		
3	Removing blank rows and replacing null values in a product dataset		
4	Calculating a formatted Display Name by combining First Name and Last Name columns		
5	Filtering a SQL Server table to only include records from the last 12 months		
6	Recording a full audit log of every change made to a dataset over time		
7	Enforcing a strict schema where any row with a wrong data type is rejected and flagged		
8	Building a non-linear data prep pipeline where some steps depend on the results of unrelated parallel branches		
9	Processing and transforming 500 million rows from a data warehouse daily		
10	A pipeline that has grown to 35 sequential transformation steps		

Hint: Review Section 3A.5 -- Power Query is great for extraction, shaping, merging, transforming, cleaning, and value manipulation. It is NOT the best tool for auditing/logging, strict type enforcement, non-linear prep, or performance/scalability at large scale.

Part 4: Short Answer (~9 minutes)

Answer each question in 2-4 sentences in your own words.

Question 1

Every month, a team manually downloads a new sales CSV, opens it in Excel, removes three columns, filters out cancelled orders, and renames headers before uploading it to a report. How does Power Query change this workflow, and which specific feature makes the improvement possible?

Question 2

A Power BI report connects to a Microsoft SQL Server database with 10 million rows. The analyst adds a filter in Power Query to only load orders from the past 90 days. Explain what query folding does in this situation and why it matters for performance.

Question 3

Three different Power BI reports in your organization all need to load and clean the same customer data from the same source. What Power Query feature would allow you to build the transformation once and share it across all three reports? Explain how it works.

Question 4

A colleague notices that Power Query Editor automatically writes code as they click through the GUI. They want to add a custom filter that the GUI cannot produce on its own. What tool would they use, and what language would they be editing?

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Bonus / Reflection (if time allows)

Section 3A notes that Power Query 'starts to break down when the number of steps exceeds 20' and is 'optimized for ease-of-use over speed.' In 3-5 sentences, describe a real-world data scenario where Power Query would NOT be the right tool and explain what you would use instead. What factors would push you toward a dedicated ETL tool like SQL Server Integration Services (SSIS)?
